FOR A MORPHISM OF FIBRED CATEGORIES THE LOWER SHRIEK FUNCTORS ON MODULES AND ABELIAN SHEAVES EXIST AND ARE COMPATIBLE, DERIVED OR NOT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the existence step quoting continuity and cocontinuity, the reduction of both commutativity assertions to the single isomorphism Lg_!^Ab j_{U'!}O_{U'} = j_{u(U')!}O_{u(U')}, the localisation to the fibres via exactness of j_! and uniqueness of adjoints, and the closing Yoneda chain of adjunction identities.

Editorial note. The label lemma-properties-lower-shriek-fibred-category names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the eleven cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Situation of a morphism of fibred categories over a base topos. Same chapter, the standing situation of the morphism g : C' -> C of fibred categories over D with its fibre functors and the morphism g' on fibres. Supplies the functors g^* = g^{-1}, the fibres C/U, C'/U' and the localised morphism g' appearing throughout.
- Existence of the lower shriek adjoint g_! and its derived version Lg_!. Modules on Sites, the sections on exactness and on lower shriek for modules, and the section on the derived lower shriek. Supplies the existence of g_!, g_!^Ab and their left derived functors, which the lemma assembles into four assertions.
- Extensionality: it suffices to test the shriek of j_{U'!}O_{U'}. Modules on Sites, the remarks characterising when the lower shriek of an extension-by-zero sheaf agrees with the shriek on the image, and the corresponding derived remark. Supplies the reduction of the two compatibility assertions to the single check on j_{U'!}O_{U'}.

Source reference. Original lemma 08PB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9795–9795. Prerequisite pointer invoked at proof line(s) 9899; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 07AE: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9436–9436. Prerequisite pointer invoked at proof line(s) 9889; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07AB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9271–9271. Prerequisite pointer invoked at proof line(s) 9875; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03EG: sources/stacks-chow/tags.txt, source lines 4455–4455. Prerequisite pointer invoked at proof line(s) 9901; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03DJ: sources/stacks-chow/tags.txt, source lines 4422–4422. Prerequisite pointer invoked at proof line(s) 9897; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 08P4: sources/stacks-chow/tags.txt, source lines 11012–11012. Prerequisite pointer invoked at proof line(s) 9895; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0798: sources/stacks-chow/tags.txt, source lines 9263–9263. Prerequisite pointer invoked at proof line(s) 9883; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 04BE: sources/stacks-chow/tags.txt, source lines 5649–5649. Prerequisite pointer invoked at proof line(s) 9872; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0796: sources/stacks-chow/tags.txt, source lines 9261–9261. Prerequisite pointer invoked at proof line(s) 9873; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08PA: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 9767–9767. Prerequisite pointer invoked at proof line(s) 9831; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 06NW: sources/stacks-chow/tags.txt, source lines 8550–8550. Prerequisite pointer invoked at proof line(s) 9868; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Assumption and notation as in Situation 08PA.*

(1) *There are left adjoints* $g_! : Mod(\mathcal{O}_{\mathcal{C}'}) \to Mod(\mathcal{O}_{\mathcal{C}})$ *and* $g_!^{Ab} : Ab(\mathcal{C}') \to Ab(\mathcal{C})$ *to* $g^* = g^{-1}$ *on modules and on abelian sheaves.*

(2) *The diagram*

$$\begin{array}{ccc} Mod(\mathcal{O}_{\mathcal{C}'}) & \xrightarrow[g_!]{} & Mod(\mathcal{O}_{\mathcal{C}}) \\ \downarrow & & \downarrow \\ Ab(\mathcal{C}') & \xrightarrow{g_!^{Ab}} & Ab(\mathcal{C}) \end{array}$$

commutes.

(3) *There are left adjoints $Lg_! : D(\mathcal{O}_{\mathcal{C}'}) \to D(\mathcal{O}_{\mathcal{C}})$ and $Lg_!^{Ab} : D(\mathcal{C}') \to D(\mathcal{C})$ to $g^* = g^{-1}$ on derived categories of modules and abelian sheaves.*

(4) *The diagram*

$$\begin{array}{ccc} D(\mathcal{O}_{\mathcal{C}'}) & \xrightarrow[Lg_!]{} & D(\mathcal{O}_{\mathcal{C}}) \\ \downarrow & & \downarrow \\ D(\mathcal{C}') & \xrightarrow{Lg_!^{Ab}} & D(\mathcal{C}) \end{array}$$

commutes.

Proof. The functor u is continuous and cocontinuous Stacks, Lemma 06NW. Hence the existence of the functors $g_!$, $g_!^{Ab}$, $Lg_!$, and $Lg_!^{Ab}$ can be found in Modules on Sites, Sections 04BE and 0796 and Section 07AB.

To prove (2) it suffices to show that the canonical map

$$g_!^{Ab} j_{U'!}\mathcal{O}_{U'} \to j_{u(U')!}\mathcal{O}_{u(U')}$$

is an isomorphism for all objects U' of $\mathcal{C}'$, see Modules on Sites, Remark 0798. Similarly, to prove (4) it suffices to show that the canonical map

$$Lg_!^{Ab} j_{U'!}\mathcal{O}_{U'} \to j_{u(U')!}\mathcal{O}_{u(U')}$$

is an isomorphism in $D(\mathcal{C})$ for all objects U' of $\mathcal{C}'$, see Remark 07AE. This will also imply the previous formula hence this is what we will show.

We will use that for a localization morphism j the functors $j_!$ and $j_!^{Ab}$ agree (see Modules on Sites, Remark 08P4) and that $j_!$ is exact (Modules on Sites, Lemma 03DJ). Let us adopt the notation of Lemma 08PB. Since $Lg_!^{Ab} \circ j_{U'!} = j_{U!} \circ L(g')_!^{Ab}$ (by commutativity of Sites, Lemma 03EG and uniqueness of adjoint functors) it suffices to prove that $L(g')_!^{Ab}\mathcal{O}_{U'} = \mathcal{O}_U$. Using the results of Lemma 08PB we have for any object E of $D(\mathcal{C}/u(U'))$ the following sequence of equalities

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{C}/U)}(L(g')_!^{Ab}\mathcal{O}_{U'}, E) &= \mathrm{Hom}_{D(\mathcal{C}'/U')}(\mathcal{O}_{U'}, (g')^{-1}E) \\ &= \mathrm{Hom}_{D(\mathcal{C}'/U')}((\pi'_{U'})^{-1}\mathcal{O}_V, (g')^{-1}E) \\ &= \mathrm{Hom}_{D(\mathcal{D}/V)}(\mathcal{O}_V, R\pi'_{U',*}(g')^{-1}E) \\ &= \mathrm{Hom}_{D(\mathcal{D}/V)}(\mathcal{O}_V, (\sigma')^{-1}(g')^{-1}E) \\ &= \mathrm{Hom}_{D(\mathcal{D}/V)}(\mathcal{O}_V, \sigma^{-1}E) \\ &= \mathrm{Hom}_{D(\mathcal{D}/V)}(\mathcal{O}_V, \pi_{U,*}E) \\ &= \mathrm{Hom}_{D(\mathcal{C}/U)}(\pi_U^{-1}\mathcal{O}_V, E) \\ &= \mathrm{Hom}_{D(\mathcal{C}/U)}(\mathcal{O}_U, E) \end{aligned}$$

By Yoneda's lemma we conclude. □